

Soybean Yield Response to Narrow Rows Is Largely due to Enhanced Early Growth

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ABSTRACT

Researchers in the southern USA have shown that the increased yield associated with narrow rows for late-planted determinant soybean [*Glycine max* (L.) Merr.] is due mainly to a greater crop growth rate during vegetative and early reproductive stages which allows for an increase in the number of fertile nodes and increased number of pods per fertile node. The objective of this study was to determine if a similar explanation is applicable to the narrow-row yield response reported with full-season narrow-row production of indeterminate soybean cultivars in the midwestern USA. To test this hypothesis field experiments were conducted in 1992 and 1993 on a Flanagan silty loam soil (fine, montmorillonitic, mesic, aquic, argiudoll, mollisol). Soybean was sown in row widths of 38, 76, and 114 cm and thinned to 450 000 plants per ha at V1 to V2. A linear decrease with increasing row width was observed for grain yield (3497, 3250, 2920 kg/ha), number of pods per plant (93, 89, 85), plant height (99, 96, 93 cm), number of branches per plant (3.3, 3.1, 2.9), and harvest index (33, 31, 30). Crop growth rate increased with decreasing row width until about R5 after which row width had no significant effect on crop growth rate. These results support our hypothesis that the yield increase noted for narrow-row indeterminate soybean production in the midwestern USA is due predominantly to beneficial effects experienced prior to the main grain-fill periods in a manner similar to that reported for late-planted narrow-row determinate soybean production in the southern USA.

SOYBEAN grain yield generally increases in the midwestern U.S. Corn Belt as row width decreases, i.e., soybean yields more when grown in narrow rows than wide rows (8,10,11,16). Producers in the midwest use mainly indeterminate soybeans of maturity groups (MG) I to IV, while soybean producers in the southern and southeastern USA (South) predominantly use determinate cultivars of MG V to VIII. The effect of row width on soybean grain yield in the South has been less consistent than that of the Midwest (1,12). Johnson et al. (9) proposed the lack of consistent response to nar-

row rows in the South is a product of the growth habit and plant type of the soybean cultivars being grown. Unless planted late, the determinate growth habit and overall size of the soybean cultivars in the South usually allows for canopy closure prior to reproductive stages even when grown in wide rows (2). When planted late, e.g., in a double-crop system, the determinate cultivars are much smaller and often (3,8,12), but not always (18), show a positive yield response to narrow rows similar to that of the Midwest.

The yield increase associated with narrow-row soybean production traditionally has been attributed to greater light interception (15,16,17). Duncan (7), however, postulated that it is not an issue of intercepting more sunlight during grain fill and that in fact wide-row systems can intercept similar amounts of incoming solar radiation and yet still yield less grain. Rather, he argued that less competition during vegetative growth stages results in greater biomass and more pods which are then able to better able to utilize the photosynthates produced with the light that is intercepted.

A number of studies have supported Duncan's postulates. Wells (19) reported that the yield difference noted between wide- and narrow-row systems in North Carolina was not explained by differences in canopy apparent photosynthesis during the grain-fill period. Similarly, Board and Harville (4), working in Louisiana, recommended a selection criteria for southern cultivars in narrow-row late plantings should concentrate on a rapid and large accumulation of leaf area index during vegetative and early reproductive stages. Parvez et al. (12), working in Florida, and Board et al. (3,5), working in Louisiana, expanded on this topic. They pointed out that narrow-row systems allow for an increase in branch yield components per unit area, greater light interception and increased crop growth rate only during vegetative and early reproductive stages. Similar increases

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Abbreviations: CGR, crop growth rate; LAI, leaf area index; LDB, leafblade dry biomass; MG, maturity group; SDB, stem dry biomass; TPDB, total plant dry biomass; t, time.

were not noted during the grain-fill periods. Both groups concluded that the increased yield associated with late-planted narrow-row systems in the South was due mainly to greater crop growth rate (CGR) during vegetative and early reproductive stages which allowed for an increase in the number of fertile nodes and increased number of pods per fertile node.

It is not known if a similar explanation is applicable to the narrow-row yield response seen with full-season narrow-row production of indeterminate soybean cultivars in the Midwest; however, it is our hypothesis that such an explanation is appropriate. The objective of this research was to test the hypothesis that soybeans produce more grain in the Midwest because of beneficial effects experienced prior to late reproductive (R5–R7) periods.

Field Procedures

Field studies were conducted during 1992 and 1993 at the Agronomy and Plant Pathology South Farm of the University of Illinois at Urbana-Champaign (40° N Lat). Conventional tillage was used. The previous crop each year was corn (*Zea mays* L.). Each year, soil testing indicated fertility was adequate for soybean production so no fertilizer was applied. The herbicides bentazon [3-(1-methylethyl)-1*H*-2,1,3-benzothiadiazin-4(3*H*)-one 2,2-dioxide; 0.84 kg a.i./ha], acifluorfen [sodium 5-(2-chloro, α , α , trifluoro-para-tolyloxy)-2-nitrobenzoate; 0.21 kg a.i./ha], and sethoxydim [2-[1-(ethoxyimino)butyl]-5-[2-(ethylthio)propyl]-3-hydroxy-2-cyclohexen-1-one; 0.21 kg a.i./ha] were applied at V3 for weed control. Hand weeding was also used.

On 18 May 1992 and 17 May 1993, the indeterminate soybean cultivar Elgin (MG II) was planted at a rate in excess of 600 000 seed/ha and then hand thinned to about 450 000 plants/ha at the V2 stage (13). The experiment was conducted in a randomized complete block design with nine replications. Individual experimental units were 138 m² (23 m long by 6 m wide). Treatments consisted of row widths of 38, 76, and 114 cm.

Sampling Procedures

Twelve to 15 sample areas, per experimental unit, were selected randomly each year. Each sample area consisted of 45 consecutive plants in one row (1 m² of land). Plastic flags were stapled to wooden stakes and used to locate sample areas throughout the growing season.

At weekly intervals, all above ground plant tissue was re-

Table 1. Estimated coefficients and their SE for the regression equations of the Logistics function [$f(x,a,b,c) = A/(1 + Be^{-cx})$] fitted to the total plant dry biomass over a 2-yr period (1992 and 1993) for 38-, 76-, and 114-cm row width treatments of Elgin soybean at Urbana, IL.

	Parameter		
	A	B	C
38-cm row width			
Parameter estimate	6.993	1.154	0.380
SE	0.160	0.187	0.072
76-cm row width			
Parameter estimate	6.935	1.235	0.384
SE	0.105	0.106	0.039
114-cm row width			
Parameter estimate	6.913	1.348	0.385
SE	0.119	0.135	0.039

moved from a single sample area within each experimental unit. Leaf area index (LAI) was estimated by measuring 20% of the leaflet blades with a LI-COR 3000 leaf-area meter (LI-COR, Inc., Lincoln, NE). Total dry biomass of leaflet blades, stem, and pods were measured after drying for 6 d in a forced air dryer at 60°C. TPDB was the sum of the individual dry matter measures. Growth stages were determined from three randomly selected plants from each sample. Stages were recorded as described by Ritchie et al. (13). At maturity, plant height, number of branches per plant, and number of pods per plant were measured for 10 plants selected randomly in each experimental unit. Grain yield, adjusted to 130 g/kg moisture, was measured by harvesting 3, 2, or 1 rows from the 38-, 76-, and 114-cm row width treatments, respectively. Yield was adjusted for the 76-cm row width to provide an estimate proportional to that of the other two row widths.

Smooth curves were fitted to the relationships between the measured plant data, total plant dry biomass (TPDB), leafblade dry biomass (LDB), and LAI; and time (t). Polynomial equations were used for LDB and LAI while the Logistics function was used for TPDB. A natural log transformation was used to obtain homoscedasticity for LDB, LAI, and TPDB.

Selection of the appropriate polynomial function for LDB, stem dry biomass (SDB), and LAI was conducted by fitting polynomial functions to each particular year-treatment combination (6). The additional sums of squares, attributed to the model by sequentially adding t, t², t³, and t⁴ were obtained for each term by the Type 1 sum of squares option of the REG procedure in the Statistical Analysis System (14). A third order

Table 2. Estimated coefficients and standard errors of polynomial regression equations for leaf area index, leaf dry biomass, and stem dry biomass over a 2-yr period (1992 and 1993) for 38-, 76-, and 114-cm row width treatments of Elgin soybean at Urbana, IL.

	Parameter			
	β_0	β_1	β_2	β_3
Leaf area index				
38-cm row width				
Parameter estimate	-1.357	1.039	-0.103	0.0026
SE	0.151	0.104	0.020	0.0011
76-cm row width				
Parameter estimate	-1.419	1.022	-0.097	0.0023
SE	0.138	0.096	0.018	0.0001
114-cm row width				
Parameter estimate	-1.570	0.974	-0.085	0.0015
SE	0.156	0.108	0.024	0.0011
Leaf dry biomass				
38-cm row width				
Parameter estimate	2.773	0.802	-0.072	0.0016
SE	0.145	0.100	0.019	0.0010
76-cm row width				
Parameter estimate	2.578	0.842	-0.072	0.0013
SE	0.144	0.099	0.019	0.0010
114-cm row width				
Parameter estimate	2.472	0.754	-0.049	-0.0001
SE	0.100	0.069	0.013	0.0001
Stem dry biomass				
38-cm row width				
Parameter estimate	1.605	1.266	-0.123	0.0039
SE	0.095	0.066	0.012	0.0007
76-cm row width				
Parameter estimate	1.601	1.175	-0.102	0.0026
SE	0.144	0.099	0.019	0.0010
114-cm row width				
Parameter estimate	1.532	1.091	-0.083	0.0015
SE	0.105	0.073	0.014	0.0008

Table 3. Growth stages and weekly means of average air temperature and total precipitation for the 1993 and 1994 growing seasons at Urbana, IL.

Weekly period	Growth stage‡	Average air temperature†		Precipitation	
		1992	1993	1992	1993
		°C		mm	
1 (22–28 June)	V3	19.4	23.6	0.11	2.05
2 (29 June–5 July)	R1	22.8	24.8	3.05	2.26
3 (6–12 July)	R2	23.5	25.1	4.59	1.14
4 (13–19 July)	R2	22.9	24.6	1.61	2.72
5 (20–26 July)	R3	22.1	23.8	2.29	1.52
6 (27 July–2 August)	R4	20.7	23.5	3.99	0.30
7 (3–9 August)	R5	27.7	19.4	0.02	0.54
8 (10–16 August)	R6	19.8	24.4	0.23	7.62
9 (17–23 August)	R6	20.2	24.7	0.14	1.53
10 (24–30 August)	R6	20.9	27.0	1.19	0.02
11 (31 August–6 September)	R7	20.9	19.5	0.08	2.78

† Average air temperature defined as the mean of the weekly mean maximum and mean minimum air temperature. Source: Atmospheric Sciences Division, Illinois State Water Survey, Champaign, IL.

‡ Growth stages as described by Ritchie et al. (1992).

polynomial was chosen for LDB, SDB, and LAI. Estimated coefficients and their standard errors for TPDB, LDB, SDB, and LAI are presented in Tables 1 and 2.

RESULTS AND DISCUSSION

In general, weather conditions during the 2 yr of this study were similar, although the 1992 growing season was drier than that of 1993 during June and August and cooler in general (Table 3). The rate of plant ontogeny was not affected by year (Table 3), but grain yield, number of pods per plant, plant height, number of branches per plant, and harvest index were all larger in 1993 than 1992 (Table 4). Grain yield, number of pods per plant, plant height, number of branches per plant, and harvest index all decreased linearly with increasing row width (Table 4). This is in agreement with Board et al. (2) who showed that in the South increased seed yield in late-planted determinate narrow-row soybeans

Table 4. Means and *F* values for seed yield, pods per plant, plant height, branches per plant, and harvest index of Elgin soybean grown in 38-, 76-, and 114-cm row widths in 1992 and 1993 at Urbana, IL.

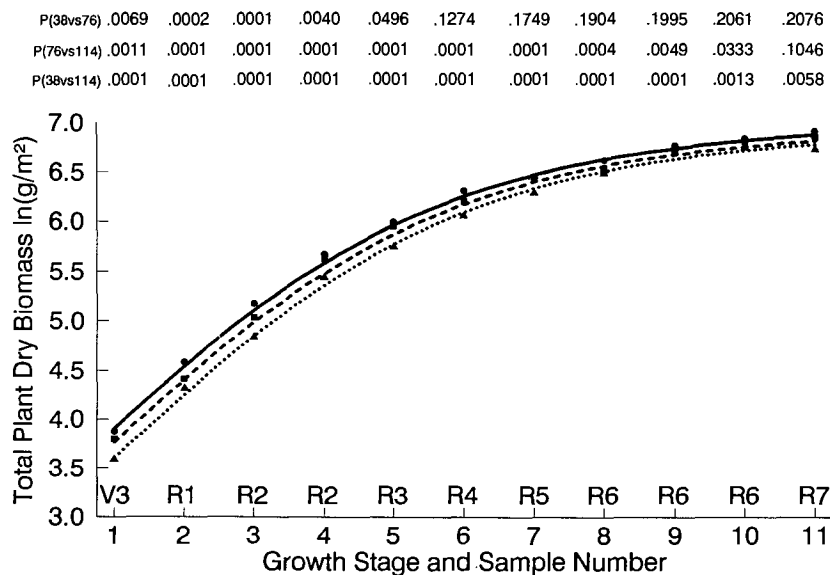
Variable	Seed yield	Pods per plant	Plant height	Branches per plant	Harvest index
	kg/ha		cm		%
Year					
1992	2998	80	90	2.9	29
1993	3447	96	101	3.3	33
Row width					
38	3497	93	99	3.3	33
76	3250	89	96	3.1	31
114	2920	85	93	2.9	30
	<i>F</i> value				
Year	21.8**	16.7**	202.4**	8.7**	7.7**
Block (year)	1.2 ns	2.8**	2.6*	1.4 ns	1.7 ns
Row	12.1**	63.1**	21.7**	4.7*	5.9**
linear	24.1**	126.0**	43.4**	9.4**	16.3**
quadratic	0.2 ns	0.3 ns	0.1 ns	0.1 ns	1.2 ns
Year × Row	0.9 ns	0.3 ns	5.0*	3.3 ns	0.1 ns

* and ** indicate significance at alpha equal to 0.01 and 0.05, respectively. ns indicates not significant at alpha equal to 0.05.

was due to increased branch biomass and seed-yield components.

With the single exception of plant height, years did not interact with the row width treatment (Table 4). In the case of plant height, the significant interaction was due to the magnitude of response rather than a difference in the shape or the direction of the response. In both years, plant height decreased linearly with increasing row width, but the slope of the response was significantly greater in 1992 ($y = 99.4 - 0.12x$) than in 1993 ($y = 104.5 - 0.04x$). Additionally, analysis of variance indicate that coefficients A (asymptote), B (intercept), and C (rate) for total plant biomass were not affected by either the main effect of year or the year × row width interaction. Based on these results all further discussion will be based upon pooled years.

With the noted exception of the 76 vs 114 cm contrast at sample number 11 (R7), the estimated TPDB of soybean grown in either the 38- or 76-cm row width was

**Fig. 1.** Total plant dry biomass curves depicting mean primary data from 2 yr (1992 and 1993) and fitted curves for Elgin soybean in 38 (●—), 76 (■----), and 114 (▲.....) cm wide rows at Urbana, IL.

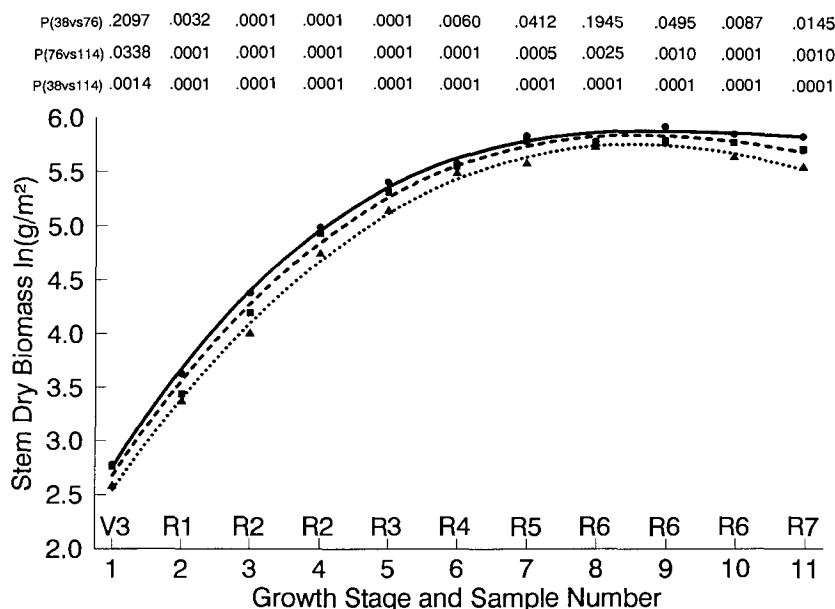


Fig. 2. Stem dry biomass curves depicting mean primary data from 2 yr (1992 and 1993) and fitted curves for Elgin soybean in 38 (●—), 76 (■---), and 114 (▲····) cm wide rows at Urbana, IL.

significantly greater than that of soybean in the 114-cm row width at all sample times (Fig. 1). However, the advantage noted with the narrower rows differed with time. For example, at sample number 1 (V3) estimated TPDM of soybean in the 38-, 76-, and 114-cm row widths was 3.87, 3.79, and 3.58 ($\ln \text{ g/m}^2$), respectively. When converted back to original units these values are approximately 48, 44, and 36 g/m^2 . Thus, at V3 soybean growing in the 38-cm row width had an estimated total plant biomass on a grams per square meter basis which was 33% larger than that of soybean grown in the widest row width. At sample time 11 (R6) the TPDB was only 17% greater.

The TPDB of soybean growing in the 38-cm row width was significantly larger than that of soybean growing in

the 76-cm row width until sample week 5 (R3), after which the two treatments were similar. Note, however, that seed yield decreased with increasing row width from 38 cm. Thus, if total plant biomass is a factor in seed yield of soybean, the factor must have played its role during the early season. This is in agreement with previous work which indicated a larger row width affect on early total dry biomass than on final plant dry biomass.

The increase in TPDB was contributed to by increases in SDB (Fig. 2) and LDB (Fig. 3). The SDB of soybean in 38- and 76-cm row widths was consistently greater than that of soybean in the 114-cm row width (Fig. 2). When the total SDB of soybean in 38- and 76-cm row widths were compared, the soybean in the 38-cm row width demonstrated a consistent numerical advantage,

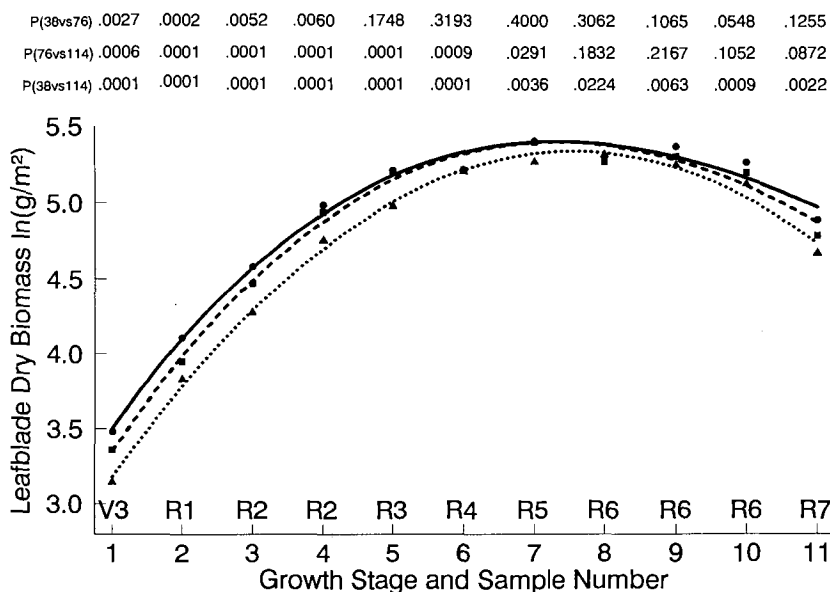


Fig. 3. Leafblade dry biomass curves depicting mean primary data from 2 yr (1992 and 1993) and fitted curves for Elgin soybean in 38 (●—), 76 (■---), and 114 (▲····) cm wide rows at Urbana, IL.

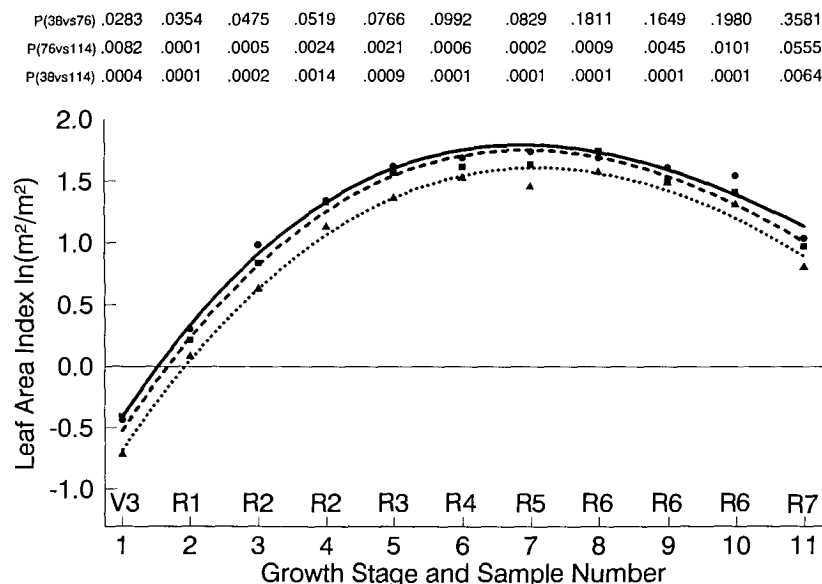


Fig. 4. Leaf area index curves depicting mean primary data from 2 yr (1992 and 1993) and fitted curves for Elgin soybean in 38 (●—), 76 (■—), and 114 (▲—) cm wide rows at Urbana, IL.

and were significantly larger at all sample times with the exception of 1 (V3) and 8 (R6). In general, the relationship is similar to that of TPDB in that differences between row width treatments were largest prior to R5.

Similarly, estimated LDB of soybean in 38-cm row widths was consistently greater than in the 114-cm row width (Fig. 3). Estimated LDB of soybean in the 76-cm row widths was numerically larger than that of soybean in the 114-cm row widths the entire season, and statistically larger prior to sample time 6 (R6). When compared with the 76-cm treatment, soybean grown in the 38-cm row width had larger LDB prior to sample time 5 (R3), but not during the main grain-fill periods.

LAI of soybean in 38- and 76-cm row widths was consistently larger than that of soybean in the 114-cm

row width (Fig. 4). The estimated LAI of soybean in 38- and 76-cm row widths were significantly different only until the sample time 4 (R2). Thus, if an increase in LAI was a contributor to the yield increase seen with the narrowest row spacing then it must be concluded that the contribution was made prior to the main grain-fill period. Southern researchers have noted that light interception was not affected by row width (4,5,19). In our study, we did not successfully measure light interception so we cannot comment on differences between the 38- and 76-cm row widths with respect to light interception other than to say that both 38- and 76-cm row width treatments resulted in complete and dense canopy cover by R5. We did note that the soybean crop grown in the 114-cm row width did not achieve canopy closure

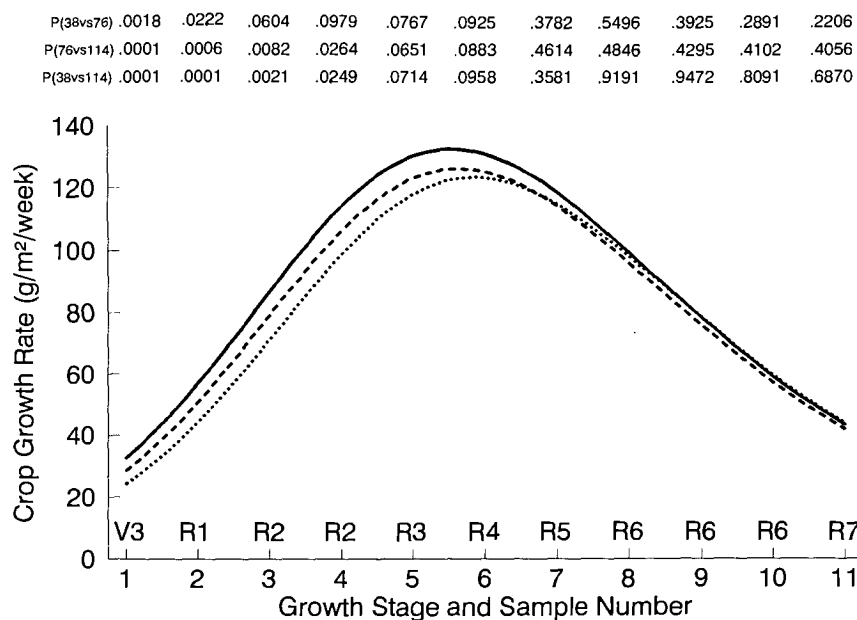


Fig. 5. Crop growth rate curves depicting derived curves for Elgin soybean in 38 (●—), 76 (■—), and 114 (▲—) cm wide rows at Urbana, IL.

at any time during either growing season. During the R5–R6 period, there was about 5 to 10 cm of soil surface in the middle of each furrow which was not covered by soybean canopy.

Crop growth rate increased with decreasing row width until about R5 ($\alpha = 0.10$), after which row width had no significant affect (Fig. 5). These results lead us to accept our hypothesis that the yield increase noted for narrow-row soybean production in the Midwest is due predominantly to beneficial effects experienced prior to the main grain-fill periods in a manner similar to that reported for late-planted narrow-row soybean production in the south. An increase in the early season LAI results in an increase in early season CGR and an increase in the number of pods. This leads us to suggest that management factors related to vigorous early season growth and development (e.g., rapid emergence, pest control etc.) are of prime importance for increasing soybean yield in both southern and midwestern soybean production areas.

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